

APPENDIX B — NON-LIMITING PARAMETERS, SAMPLING MODES, AND QUALITY INDICES

A. Overview

This appendix provides non-limiting examples of parameters, sampling modes, and quality index constructions that may be used in any embodiment described herein. Values and ranges are illustrative and may be adjusted based on device constraints, user comfort, and sensor modality substitution.

B. Sampling Modes (Non-Limiting)

Any node may operate in one or more sampling modes, including: 1. Continuous sampling for longitudinal pattern capture. 2. Periodic sampling with duty cycling to reduce power consumption. 3. Event-triggered sampling based on thresholds, detected state transitions, or user-initiated actions. 4. Burst sampling in which high-rate samples are collected during validated windows (e.g., low motion or stable contact).

C. Example Sampling Rate Ranges (Non-Limiting)

Sampling rates may vary by modality, including non-limiting examples: Pressure/strain/magnetic displacement: about 1–500 Hz, or higher as needed for dynamic events. PPG / optical pulse: about 10–500 Hz depending on wavelength count and power constraints. Impedance / multi-frequency impedance: about 0.1–50 Hz for slow trend sensing, and/or burst modes for spectra acquisition. pH sensing: about 0.01–10 Hz depending on sensor dynamics and drift compensation. IMU: about 5–1,000 Hz depending on motion gating requirements. Time-of-flight / lidar / IR distance: about 1–200 Hz depending on geometry acquisition and power.

D. Windowing and Segmentation (Non-Limiting)

The computing system may segment data into windows comprising: fixed windows (e.g., 10 seconds, 30 seconds, 1 minute, 5 minutes), rolling windows with overlap, and/or event-centered windows around detected transitions (e.g., onset/offset events). Windows may be labeled using context and validity tags, including sleep proxies, activity state, posture proxies, thermal context tags, and/or wear-integrity tags.

E. Signal Conditioning and Normalization (Non-Limiting)

Signal conditioning may include one or more of: 1. filtering (low-pass, band-pass, notch), 2. baseline removal and drift compensation, 3. temperature compensation, 4. sensor fusion smoothing, 5. resampling and interpolation to common time bases, and 6. normalization to within-user baselines and/or geometry-derived references.

F. Quality Index Construction (Non-Limiting)

In various embodiments, a quality index (QI) or confidence indicator is computed using one or more of: motion state (IMU magnitude, jerk, orientation stability), contact integrity (pressure/contact baselines, capacitive contact confirmation, electrode impedance stability, optical coupling checks), sensor saturation/limits (clipping detection, out-of-range detection), multi-sensor agreement (e.g., pressure + strain + hall-effect consistency), temporal continuity (dropout rate, discontinuities), and environmental plausibility (temperature discontinuities, moisture outliers). Quality indices may be computed as a scalar score (e.g., 0–1), as discrete labels (e.g., Valid / Reduced Confidence / Invalid), or as multi-component

confidence vectors (e.g., Motion Confidence, Contact Confidence, Physiologic Confidence).

G. Example Decision Rules (Non-Limiting)

The system may apply one or more non-limiting decision rules, including: 1. Reject windows with motion above a threshold unless corroborated by stable contact integrity. 2. Down-weight windows with reduced optical coupling or electrode impedance instability. 3. Require multi-sensor agreement before declaring a detected pattern event. 4. Prefer context-matched windows (e.g., low-motion sleep windows) for correlation scoring.

H. Calibration and Self-Checks (Non-Limiting)

Nodes may perform self-checks and calibration routines including: sensor offset calibration, temperature compensation calibration, optical coupling verification, electrode impedance verification, contact confirmation checks, and scan-path validation checks for session-based geometry acquisition.

I. Variations

Any parameter, sampling mode, and QI method described herein may be altered, substituted, combined, or omitted while maintaining the functional intent of generating privacy-preserving metrics with confidence labeling and supporting cross-device and cross-user correlation under consent gating.